Creating a tridiagonal matrix

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Aaron Atkinson on 11 Nov 2019   Vote 0   Link   Flag

Commented: John D'Errico LEVEL 9 MVP on 10 Dec 2022

Accepted Answer: Stephen23 LEVEL 10 MVP

I am currently trying to create a 500*500 matrix in matlab with diagonals a=-1, b=4, c=2. My teacher has said that the best way to go about it is using loops, but is there a coded in function to use?

2 Comments

David Goodmanson LEVEL 7 EDITOR on 11 Nov 2019

Hi Aaron

check out the 'diag' function

Alex Treat on 30 Oct 2020

coughs you were in the mec 103 class at CSU...

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Accepted Answer

Stephen23 LEVEL 10 MVP
on 11 Nov 2019   Vote 11   Link   Flag

Edited: Stephen23 LEVEL 10 MVP on 20 Mar 2022

"My teacher has said that the best way to go about it is using loops"

Why on earth would they say that? Here are some non-loop aproaches:

1- See @giannit's comment: https://www.mathworks.com/matlabcentral/answers/490368-creating-a-tridiagonal-matrix#comment_1027546

2- Use `diag` :

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```
>> N = 10;
>> a = -1;
>> b = 4;
>> c = 2;
>> M = diag(a*ones(1,N)) + diag(b*ones(1,N-1),1) + diag(c*ones(1,N-1),-1)
M =
  -1   4   0   0   0   0   0   0   0   0
   2  -1   4   0   0   0   0   0   0   0
   0   2  -1   4   0   0   0   0   0   0
   0   0   2  -1   4   0   0   0   0   0
   0   0   0   2  -1   4   0   0   0   0
   0   0   0   0   2  -1   4   0   0   0
```

```
    0    0    0    0    0    2   -1    4    0    0
    0    0    0    0    0    0    2   -1    4    0
    0    0    0    0    0    0    0    2   -1    4
    0    0    0    0    0    0    0    0    2   -1
```

3- indexing is reasonably simple:

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```
>> M = zeros(N,N);
>> M(  1:1+N:N*N) = a;
>> M(N+1:1+N:N*N) = b;
>> M(  2:1+N:N*N-N) = c
M =
   -1    4    0    0    0    0    0    0    0    0
    2   -1    4    0    0    0    0    0    0    0
    0    2   -1    4    0    0    0    0    0    0
    0    0    2   -1    4    0    0    0    0    0
    0    0    0    2   -1    4    0    0    0    0
    0    0    0    0    2   -1    4    0    0    0
    0    0    0    0    0    2   -1    4    0    0
    0    0    0    0    0    0    2   -1    4    0
    0    0    0    0    0    0    0    2   -1    4
    0    0    0    0    0    0    0    0    2   -1
```

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AP

Arth Patel on 29 Sep 2020

Can you please explain the second method a bit ? It's not clear to me how you're indexing a matrix using just one argument.

SC

Stephen23 LEVEL 10 MVP on 30 Oct 2020

"It's not clear to me how you're indexing a matrix using just one argument."

The second example uses *linear indexing*:

https://www.mathworks.com/help/matlab/math/array-indexing.html

https://www.mathworks.com/company/newsletters/articles/matrix-indexing-in-matlab.html

SD

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More Answers (1)

JB

Jihen on 10 Dec 2022

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```
function[A]=remplissage(n)
R1=(-4)*ones(n-1,1);
R2=ones(n-2,1);
A=6*eye(n)+diag(R1,-1)+diag(R1,1)+diag(R2,2)+diag(R2,-2);
end
```

1 Comment

John D'Errico LEVEL 9 MVP on 10 Dec 2022

This does not actually answer the question, creating instead a matrix with 5 diagonals, so a penta-diagonal matrix.

SD

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